gateTunnelConnection

Service Description

Service ID: *"gateTunnelConnection"*

Abstract
This document describes the `gateTunnelConnection` service, version 5.2.2, produced by the GateTunnel support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `gateTunnelConnection` service, which provides relay sessions between the local cloud and a neighbour cloud: provider and consumer sessions opened by the Gatekeeper, and the legacy control sessions. It is produced by the GateTunnel system [1] in service version 5.2.2.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

Successor of the Arrowhead 4.6 Gateway (`/gateway` paths are still accepted).

1.2 How This Service Is Meant to Be Used

The local Gatekeeper opens a provider session (`open-provider-session`) when a neighbour's consumer negotiated one of our providers, and a consumer session (`open-consumer-session`) for a local consumer; the consumer session listens on a local port, and every request is forwarded over the relay. `connect-provider`, `connect-consumer` and `close-sessions` keep the v4 control-session API.

1.3 Important Delimitations and Dependencies

Only the Gatekeeper opens relay sessions. Sessions are restricted to the negotiated peer, consumer and methods/paths; default lifetimes 12 h maximum and 1 h idle. Consumer sessions are volatile and renegotiated after restart. The legacy control sessions have no data plane.

2 Service Interface

This section describes the interfaces to the `gateTunnelConnection` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

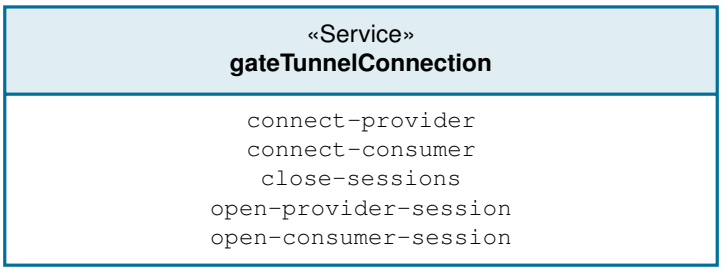


Figure 1: SysML-style block of the `gateTunnelConnection` service and its operations.

The following interface operations are available.

2.1 operation connect-provider (ControlSessionRequest) : ControlSessionResult

Creates a legacy control session for a provider (v4 API).

2.2 operation connect-consumer (ConnectConsumerRequest) : ControlSessionResult

Connects a consumer to a legacy control session.

2.3 operation close-sessions (ClosePortsRequest) : List<Map<String, Object>>

Closes legacy control sessions by port; returns the ports that failed.

2.4 operation open-provider-session (ProviderSessionRequest) : SessionEndpoint

Gatekeeper only: opens a relay session towards a local provider for a negotiated foreign consumer.

2.5 operation open-consumer-session (ConsumerSessionRequest) : SessionEndpoint

Gatekeeper only: opens a local listener that forwards a local consumer's requests to the provider cloud.

3 Information Model

Here, all data objects that can be part of the `gateTunnelConnection` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.8, which are used to represent things like hashes and identifiers.

3.1 struct ClosePortsRequest

Field	Type	Description
ports	List<Number>	Control ports to close (a bare JSON array is also accepted).

3.2 struct ConnectConsumerRequest

Field	Type	Description
queueId	Identifier	Relay queue of the provider session.

3.3 struct ConsumerSessionRequest

Field	Type	Description
sessionId	Identifier	Session identifier from the provider cloud.
consumer	Name	Consumer system.
providerCloud	String	Provider cloud.
provider	Name	Provider system.
serviceDefinition	Name	Service definition.
peer	String	Peer GateTunnel common name.

3.4 struct ControlSessionRequest

Field	Type	Description
consumer	Map<String, Object>	Consumer system descriptor.
provider	Map<String, Object>	Provider system descriptor.
consumerCloud	Map<String, Object>	Consumer cloud.
providerCloud	Map<String, Object>	Provider cloud.
relay	Map<String, Object>	Relay descriptor.
serviceDefinition	Name	Service definition.
queueId	Identifier	Relay queue; generated if absent.
peerName	String	Peer name; generated if absent.

3.5 struct ControlSessionResult

Field	Type	Description
sessionId	Identifier	Session identifier.
queueId	Identifier	Relay queue.
peerName	String	Peer name.
serverPort	Number	Local control port.
state	String	Session state.

3.6 struct ProviderSessionRequest

Field	Type	Description
sessionId	Identifier	Session identifier.
consumer	Name	Consumer system.
consumerCloud	Name	Consumer cloud, must match the peer certificate.
provider	Name	Provider system.
serviceDefinition	Name	Service definition.
peer	String	Peer GateTunnel common name.
target	Map<String, Object>	Provider endpoint: `address`, `port`, `basePath` and the negotiated `operations`.

3.7 struct SessionEndpoint

Field	Type	Description
address	String	Advertised address of the local session listener.
port	Number	Port of the session listener.
sessionId	Identifier	Session identifier.

3.8 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.

4 References

[1] J. Delsing, “GateTunnel – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-GateTunnel-java-spring/documentation/v5.2.2/GateTunnel-v5.2.2-SysD.pdf.

[2] ——, “gateTunnelConnection HTTP/TLS/JSON – Interface Design Description (GateTunnel),” Eclipse Arrowhead, IDD 5.2.2, 2026, gateTunnel-v5.2.2-gateTunnelConnection-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	